Multiplication by y exchanges two packets

Use $k[x] \oplus yk[x]$ in $k[x,y]/(y^2 - x^3 + x)$.

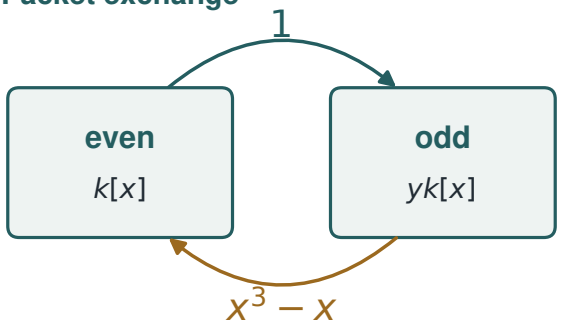
Weighted transfer blocks

$$M_y = \begin{bmatrix} \text{even} & \text{odd} \\ 0 & x^3 - x \\ 1 & 0 \end{bmatrix}$$

Columns are inputs.

Rows are outputs.

Packet exchange



$$M_y^2 = (x^3 - x)I$$

Two applications preserve each packet.